USAGE & OPPORTUNITY MODELING

A mathematical and implementation framework for predicting NBA, NFL and NCAAF player statistics

Strategy 05 | Player-stat forecasting | Quantitative methodology documentation

A hierarchical framework that separately forecasts game environment, player participation, opportunity allocation and efficiency - then converts them into complete player-stat probability distributions.

1. Executive Summary

A player’s final production is driven by how many opportunities they receive and how efficiently they convert those opportunities. Separating the two helps distinguish sustainable changes in role from noisy short-term outcomes.

$$\text{Production} = \text{Opportunity} \times \text{Efficiency}$$

Sport	Opportunity drivers	Markets
NBA	Minutes, possessions, usage, FGA, rebound chances, potential assists	Points, rebounds, assists, 3PM, steals, blocks, turnovers, PRA
NFL	Plays, dropbacks, snaps, routes, targets, carries	Receiving/rushing yards, receptions, attempts, passing yards, TDs
NCAAF	Pace, offensive scheme, play shares, dual-threat QB role	Passing, rushing and receiving player props

2. Mathematical Foundation

Let N denote opportunity count, Y(j) the contribution from opportunity j, and X the game-level statistic. The fundamental random-sum representation is:

$$X = \Sigma[j = 1 \text{ to } N]\ Y(j)$$

If conditional efficiency is constant across opportunity counts and game conditions, the familiar product-of-means equation follows. In general, preserve dependence by conditioning on context Z.

$$E[X] = E[N]\ E[Y] \text{ (under suitable assumptions)}$$

$$E[X] = E[\ N \times E(Y \mid N, Z)\ ] \text{ (general formulation)}$$

Under independent identically distributed contributions and an independent random count, variance separates into opportunity uncertainty and conversion uncertainty:

$$\text{Var}(X) = E[N]\ \text{Var}(Y) + \text{Var}(N)\ (E[Y])^2$$

For a dependent hierarchical model, use Var(X) = E[Var(X | N,Z)] + Var(E[X | N,Z]).

Hierarchical generation

1 Game environment (pace, expected plays, matchup and game script) → **2 Participation** (minutes, snaps and routes) → **3 Allocation** (usage, targets and attempts) → **4 Efficiency** (conversion and per-event production) → **5 Predictive distribution**.

3. NBA Model

3.1 Playing time and possessions

Use recency-weighted minutes, then adjust for injuries, rotation, foul risk, matchup and blowout scenarios. Regulation team minutes obey Σ player minutes = 240; overtime is modeled separately.

$$\text{Projected minutes} = \Sigma[t]\ \lambda^{(t-1)}\ M(t) \ / \ \Sigma[t]\ \lambda^{(t-1)} + \text{role adjustments}$$

$$\text{Player on-court possessions} \approx \text{expected pace} \times \text{projected minutes} / 48$$

For example, 34 minutes at pace 102 implies approximately 72.25 on-court offensive team possessions. A minutes scenario mixture is preferable if the player might start or come off the bench.

3.2 Usage and scoring

Usage rate approximates the share of team possessions a player finishes with a field-goal attempt, free-throw sequence or turnover. The coefficient 0.44 is an approximation for free-throw possession consumption.

$$\text{USG} \approx [(\text{FGA} + 0.44\ \text{FTA} + \text{TOV})\ (\text{TeamMin} / 5)] \ / \ [\text{Min}\ (\text{TeamFGA} + 0.44\ \text{TeamFTA} + \text{TeamTOV})]$$

$$\text{Expected finished possessions} \approx \text{expected on-court possessions} \times \text{usage rate}$$

For points, explicitly model two-point attempts, three-point attempts and free throws; conditional made counts can initially use binomial distributions:

$$E[\text{PTS}] = 2\ E[\text{2PA}]\ p(\text{2P}) + 3\ E[\text{3PA}]\ p(\text{3P}) + E[\text{FTA}]\ p(\text{FT})$$

Example: 10 two-point attempts at 55%, seven threes at 38%, and six free throws at 85% produce 2(10)(.55) + 3(7)(.38) + 6(.85) = **24.08 projected points**. Model FGA, shot mix and FTA jointly rather than mechanically treating usage as FGA share.

3.3 Rebounds, assists and additional markets

$$E[\text{REB}] = E[\text{offensive rebound chances}]\ p(\text{OREB}) + E[\text{defensive rebound chances}]\ p(\text{DREB})$$

$$E[\text{AST}] = E[\text{potential assists}]\ p(\text{teammate conversion})$$

Rebound chances depend on reboundable misses, on-court participation, location and lineup competition. Potential assists depend on ball-handling role, minutes, teammate shot selection and shot-making. Three-pointers made follow attempt volume × 3P conversion; steals and blocks use eligible defensive events × event-conversion probability. For PRA, simulate the underlying statistics jointly to preserve covariance.

4. NFL Model

4.1 Team plays, game script and passing volume

$$\text{Expected pass plays} \approx \text{expected offensive plays} \times \text{pass-play probability}$$

$$\text{Expected run plays} \approx \text{expected offensive plays} \times (1 - \text{pass-play probability})$$

Project team play count from both teams' pace, expected possession time, scoring environment and game script. Make pass probability conditional on score differential and clock. Keep sacks, scrambles, penalties and kneel-downs consistent with your data definitions.

4.2 Wide receiver targets, catches and yards

$$\mathrm{E}[\text{Targets}] \approx \mathrm{E}[\text{Team dropbacks}] \times \text{route participation} \times \text{targets per route}$$

$$\mathrm{E}[\text{Receptions}] \approx \mathrm{E}[\text{Targets}] \times \text{catch probability}$$

$$\mathrm{E}[\text{Receiving yards}] \approx \mathrm{E}[\text{Targets}] \times \text{catch probability} \times \mathrm{E}[\text{yards} \mid \text{catch}]$$

Example: 40 team dropbacks × 90% route participation × 25% targets per route = nine targets. At 68% catch probability and 12.5 yards per catch, forecast 6.12 receptions and **76.50 receiving yards**. In the full simulation, sample each layer conditionally and optionally separate target depth, completed air yards and yards after catch.

4.3 Rushing, passing and touchdowns

$$\mathrm{E}[\text{RB carries}] = \mathrm{E}[\text{eligible team RB carries}] \times \text{player carry share}$$

$$\mathrm{E}[\text{Rushing yards}] \approx \mathrm{E}[\text{Carries}] \times \mathrm{E}[\text{yards per carry}]$$

For illustration, 27 eligible RB carries × 70% share × 4.4 yards/carry = **83.16 expected rushing yards**. Model explosive gains with a mixture distribution. QB completions depend on attempts × completion probability; passing yards benefit from short/intermediate/deep attempt splits.

$$\mathrm{E}[\text{QB pass yards}] \approx \Sigma[\text{depth bucket } k]\ \mathrm{E}[\text{attempts}(k)]\ p(\text{complete} \mid k)\ \mathrm{E}[\text{yards} \mid \text{completion}, k]$$

Touchdowns require an explicit red-zone or goal-line opportunity model. As a rough baseline, TD count can be Poisson(λ), giving P(at least one TD) = $1 - \exp(-\lambda)$. For $\lambda = .55$, this is approximately 42.31%; possession-level simulations capture dependencies more coherently.

5. NCAAF Model

College models reuse the NFL architecture with wider variation in offensive scheme, pace, opponent strength, participation, and blowout risk. Use team-specific priors and hierarchical shrinkage to manage less consistent historical coverage.

$$\text{Expected team plays} = \beta_0 + \beta_1\,\text{TeamPace} + \beta_2\,\text{OppPace} + \beta_3\,\text{Spread} + \beta_4\,\text{Total} + \beta_5\,\text{Scheme}$$

5.1 Dual-threat quarterbacks

$$\text{QB total yards} = \text{passing yards} + \text{rushing yards}$$

$$\text{QB rush yards} = \text{designed-run yards} + \text{scramble yards} - \text{sack yards lost}$$

Apply the official data and settlement convention: NCAA sacks generally reduce individual and team rushing totals. Account for designed QB runs, scrambles, sacks and kneel-downs as separately modeled opportunity types.

5.2 Competition strength

Raw production against mismatched opponents is a poor standalone input. One hierarchical formulation is log(expected production) = player effect + team-scheme effect + opponent effect + game-context terms. Partial pooling protects small-sample player and opponent estimates.

6. Estimation and Calibration

6.1 Weighted history and prior shrinkage

$$\text{Weighted metric} = \Sigma[t]\, w(t)\, X(t) \,/\, \Sigma[t]\, w(t),\ w(t) = \exp(-k(t-1))$$

Opportunity measures may use faster decay when role changes; efficiency rates often benefit from longer histories. For a binary conversion parameter p, use a Beta prior and a binomial likelihood:

$$p \mid \text{data} \sim \text{Beta}(\alpha 0 + \text{successes},\ \beta 0 + \text{failures})$$

$$E[p \mid \text{data}] = (\alpha 0 + \text{successes}) \,/\, (\alpha 0 + \beta 0 + \text{trials})$$

Example: Beta(14,6) prior and 18 catches on 20 targets yields a posterior mean of (14+18)/(14+6+20) = **80%**, rather than the raw 90%.

6.2 Distribution choices

Quantity	Starting distribution	Consideration
Counts: targets, attempts, shot volume	Poisson / negative binomial	Negative binomial permits overdispersion
Player share of finite team volume	Binomial / multinomial	Respects coherent team-level allocation
Makes or receptions, conditional on attempts	Binomial	Allow context-dependent conversion rates
Yards per event	Empirical / mixture / hierarchical	Capture skew and explosive-play tail
Playing time	Truncated distribution / scenario mixture	Respect physical minutes and rotation constraints

7. Monte Carlo Simulation

For every iteration, sample a common game state, team opportunities, individual participation, allocation and efficiency. Sharing team-level states keeps teammate outputs coherent and preserves QB-receiver, usage-minutes and game-script dependencies.

$$\text{Simulation } s\text{: environment} \rightarrow \text{participation} \rightarrow \text{opportunities} \rightarrow \text{conversions} \rightarrow X(s)$$

$$\text{Mean} \approx \Sigma[s]\, X(s) \,/\, S;\ P(X > L) \approx \Sigma[s]\, 1\{X(s) > L\} \,/\, S$$

Retain the empirical cumulative distribution $F(x) = P(X \le x)$, percentile bands, probability mass by stat value and line-specific over/under/push probabilities. On whole-number lines, pushes may have positive probability.

8. Market Evaluation

$$\text{Implied P at } -a \text{ odds} = a / (a + 100); \text{ at } +b \text{ odds} = 100 / (b + 100)$$

$$\text{Expected value per \$1 staked} = P(\text{win}) \times \text{net payout per \$1} - P(\text{lose})$$

For -110 odds, win profit per $1 is 100/110 = .909091, and break-even win probability with no pushes is 52.380952%. At a model win probability of 58% and no push: EV = .58(.909091) − .42 = **.107273** per dollar (about 10.73%), conditional on model calibration.

Distinguish model mean, win probability and expected value. Parameter uncertainty may be assessed using posterior simulations, calibrated probability bands and P(EV > 0 | data). A projected mean above the line does not in itself prove positive value.

9. Application Architecture

Pipeline: data ingestion → normalized player/game IDs → engineered opportunity and efficiency features → model fitting → shared-game simulation → probability and market evaluation → Dash dashboard and diagnostics.

Module	Responsibility
opportunity_model.py	Minutes, snaps, plays, routes, touches, targets and attempts
efficiency_model.py	Conversion rate estimation and context adjustments
nba_model.py / nfl_model.py / ncaaf_model.py	Sport-specific generation and settlement conventions
simulation.py / distributions.py	Correlated outcome samples, cumulative probabilities and intervals
market_comparison.py	Line-specific win, loss, push and expected value
backtesting.py / calibration.py	Walk-forward validation and probabilistic diagnostics

Standard outputs: projected_mean, projected_median, projected_std, opportunity_projection, efficiency_projection, over_probability, under_probability, push_probability and estimated_ev.

10. Walk-forward Backtesting

For every historical game, use only data available as of the intended forecast timestamp, save the predictive distribution and contemporary market prices, then evaluate against the final box score. Never incorporate injury announcements, line movements or results that occurred later.

$$\text{MAE} = (1/n)\ \Sigma\ |\text{actual} - \text{forecast}|; \text{RMSE} = \text{sqrt}((1/n)\ \Sigma\ (\text{actual} - \text{forecast})^2)$$

$$\text{Brier score} = (1/n)\ \Sigma\ (\text{predicted probability} - \text{outcome indicator})^2$$

Check calibration of over/under probabilities and predictive intervals by sport, market, player role and opportunity volume. Examine error by component rather than only the end statistic: participation error, allocation error, conversion error and simulation error. Treat pushes separately or evaluate a defined three-outcome forecast.

11. Dash Dashboard Recommendation

Build a dedicated Usage & Opportunity Analytics page with player projection cards, participation and opportunity trends, efficiency trends, decomposition of expected production, histogram and cumulative distribution, market comparison, and model-validation views.

An interactive scenario panel can change minutes, usage, route participation, target share, game pace or teammate availability and regenerate the downstream projected distribution, with coherent reallocation of team opportunities.

12. Development Roadmap

Phase	Objective	Deliverable
1	Historical opportunities	Baseline player participation and opportunity forecast
2	Efficiencies and shrinkage	Context-aware player production mean
3	Matchup and role effects	Injury, rotation and game-script adjustments
4	Correlated Monte Carlo	Complete predictive distributions
5	Historical market integration	Line probabilities, EV and push handling
6	Walk-forward evaluation	Calibration and accuracy diagnostics
7	Dashboard delivery	Explorable player and scenario analytics

13. Final Takeaway

Usage & Opportunity Modeling is strongest when it predicts the mechanisms of player production rather than merely extrapolating box-score results. Model game environment, participation and allocation first; estimate efficiency with suitable shrinkage; preserve conditional dependence in simulation; then assess the calibrated distribution against each available prop line. This framework is reusable across NBA, NFL and NCAAF while respecting each sport’s statistical conventions.